



GROWTERN
Academy

45-DAYS SUMMER INTERNSHIP ON AI-POWERED **DATA ANALYTICS**

MODULE : 01 **FUNDAMENTALS OF DATA ANALYTICS**

- Introduction to data analytics & lifecycle
 - Types of data (structured/unstructured)
 - Basics of Excel (formulas, functions)
 - Data cleaning & preprocessing
 - Mini Task: Data cleaning in Excel
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MODULE : 02 **ADVANCED EXCEL & DATA HANDLING**

- Advanced formulas (VLOOKUP, HLOOKUP, IF)
 - Pivot tables & charts
 - Data visualization basics
 - Introduction to statistics (mean, median, mode)
 - Mini Project: Sales data analysis in Excel
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MODULE : 03 **SQL FOR DATA ANALYSIS**

- Introduction to databases
- SQL basics (SELECT, WHERE, ORDER BY)
- Joins, grouping, aggregation functions
- Data extraction & manipulation
- Practice queries & tasks





MODULE : 04 PYTHON FOR DATA ANALYTICS

- Introduction to Python
- Libraries: NumPy, Pandas, Matplotlib
- Data analysis using Python
- Data visualization techniques
- Mini Project: Data analysis using Python

MODULE : 05 DASHBOARD & FINAL PROJECTS

- Introduction to Power BI / Tableau
- Data visualization dashboards
- **Real-world projects:**
 - **E-Commerce Sales Dashboard**
 - **Sales Data Analysis**
 - **Stock Market Dashboard**
- Final presentation & evaluation

AI TRAINING MODULE : GENERATIVE AI & AI TOOLS FOR DEVELOPERS

- Introduction to Generative AI and its role in modern data analytics
- Introduction to AI tools: ChatGPT, Google Gemini, Claude, GitHub Copilot, Perplexity, ElevenLabs, SuperBot
- Prompt engineering for data analysis and insights generation
- Data analysis using AI tools (data cleaning, visualization, insights)
- Building a mini data analytics project using AI-assisted workflow
- Understanding limitations, ethics, and best practices of AI in data analytics

OUTCOME

Students will gain strong data analytics fundamentals, develop skills in Excel, SQL, Python, and data visualization, and build real-world data-driven projects.

